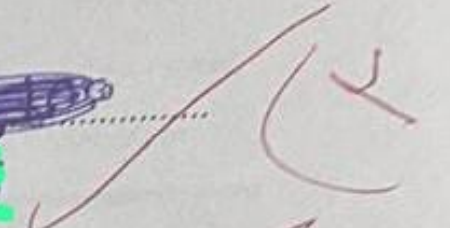


تمارين على Taylor and Maclaurin Series

✓ A-Given that $f(2) = 6$, $f'(2) = -\frac{1}{2}$ and $f''(2) = 10$.

The most accurate Taylor polynomial approximation of $f(2.2)$ is...



2. The Taylor Series is used to approximate functions using _____ and its derivatives.

e.g: Using Taylor polynomial of order (2), 5

approximate $\sin\left(\frac{\pi}{6}\right)$ taking $x_1 = \frac{\pi}{4}$.

Then compute E_t^0 .

e.g.: use Taylor polynomial of 1 term through 3 terms to approximate $\ln 2$, take 1 as a base value?

newton-rafson method تمایزین علی

2- Applying Newton-Raphson method to estimate the root of $\sqrt{7}$ up to two decimal places
Starting with the initial guess $x_0 = 2$, the first iteration x_1 is:

(A) 1.25

(B) 2.88

(C) 2.75

(D) 2.67

(D) -88

a) singular

5. The equation $f(x)$ is given as $x^3 + 4x + 1 = 0$. Considering the initial approximation at $x_0 = 1$ then the approximated value of x_1 , applying Newton Raphson method to find the root is:

a) 0.15886

b) 0.11286

c) 0.12936

d) 0.14286

Q1- Find the root of the function: $f(x) = \sin(x) + x^2 + x - 10$ using: (12 marks)

(a) ~~.....~~

(b) The *Newton Raphson Method*. Perform two iterations to find the root, assume $x_0 = 2$.

$$x_2 = \dots\dots\dots$$

Note: work to 5 significant figures

تمارين على bisection method

Q1- Find the root of the function: $f(x) = \sin(x) + x^2 + x - 10$ using: (12 marks)

(a) The *Bisection Method*. Perform two iterations to find the root, use the interval (2, 4)

$$x_{r3} = \dots\dots\dots$$

a) 11

b) 29

☒ c) 45

2. If the function $f(x) = e^x - 4$ has a root within the interval $x_0 = 1$ and $x_1 = 2$, what is the next value of the approximation of (x_{r1}) after (x_{r0}) using the bisection method.

a) 0.745

b) 1.75

☒ c) 1.25

d) 1.44

with the true value, but do not agree with

(B) $[-1, 0.2]$

(C) $[0.1, 0.4]$

7- If the function studied by bisection method changes sign between the initial values 1 and 4 and we know that an error less than 0.00583 is acceptable. Then the required number of iterations that will produce the desired result is:

(A) 8

(B) 12

(C) 14



على_الماتركس وعمل_Sedal_Gause_&_elemation

Given:

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 20.3 \\ 11.3 & -9 & -11.3 \\ 10.3 & -7 & 8.5 \end{bmatrix}, [B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 12 & -20 \end{bmatrix} \text{ then if } [C] = [A][B], \text{ then } C_{31} =$$

Answer:

The lower triangular matrix $[L]$ in the $[L][U]$ decomposition is

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 1 & 0 \end{bmatrix}$$

4- Consider the following system of linear equation $3x+y-z=12$, $x+5y+2z=14$, $x+4y+9z=10$. If the current approximation is $x=4$, $y=2$, $z=0.889$, then the Gauss-Seidal method will give next approximation as $x=3.63$, $y=1.718$ and $z=$

(A) 1.055 (B) 1.014 (C) 0.930 (D) 0.844

Solve the following systems of linear equations

(a) Use the *Cramer's Rule* to solve the following two simultaneous equations, determine x_1 and x_2 :

$$x_1 + x_2 = 25$$

$$6x_1 - 3x_2 = 6$$

$$x_1 = \dots\dots\dots, x_2 = \dots\dots\dots$$

Use the following three simultaneous equations.

(b) Use the *Gauss Elimination Method* to solve the following three simultaneous equations.
Determine the values of x, y, z :

$$-3x - y + 2z = 25$$

$$4x + 2y + z = 16$$

$$5x + 3y - \frac{z}{2} = 3$$

$x = \dots\dots\dots$, $y = \dots\dots\dots$, $z = \dots\dots\dots$

C-The value of α so that the linear system represented by the following augmented matrix has infinitely many solutions?

$$\left[\begin{array}{cc|c} 2 & 6 & 8 \\ 1 & \alpha & 4 \end{array} \right]$$

The value of α is.....

D-The system of linear equations is represented by the following augmented matrix as shown:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array} \right].$$

The value of cofactor a_{32} is.....

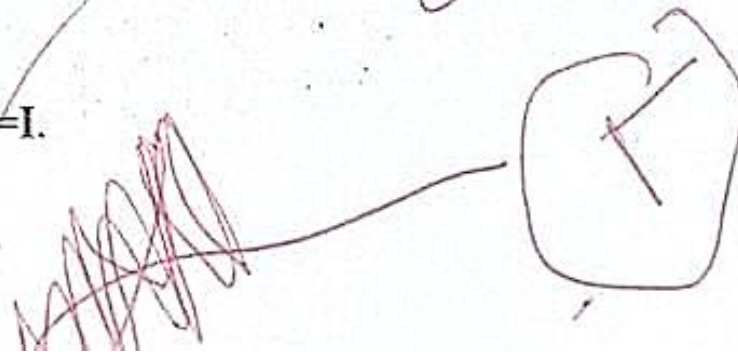
✓ F- At the end of Gauss Elimination steps on the 3×3 system of linear equations written in augmented matrix form as shown below:

$$\left[\begin{array}{ccc|c} 5 & 0 & -1 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 2 & 4 \end{array} \right]$$

The determinant of the original coefficient matrix is.....10

✓ G-If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, and $A + A^T = I$.

The value of α is.....1,047.....



اسئلة على اول موضوع (Error)

Eg :

	Rivet	Bridge
True length	10 cm	10 000 cm
Approximate	7 cm	9999 cm

Find the abs. true error and relative true error for both the rivet and the bridge.

e.g: If the true weight of car is 1000 Kgs
and the approximation was 999 Kgs.



Also, the ^{true} weight of a boy 20 Kgs
and the approximation was 19 Kgs.

Find the true error and the true percent
relative error for the car and the boy.

3- The relative error in an approximate solution is 0.004%. The number of significant digits in the solution that we can trust at least is:

(A) 5

(B) 3

☒ 4

(D) 2

(U) 1.67

e.g: $X^{\text{current}} = 5.5$, $X^{\text{previous}} = 3.5$

find E_a

Bairstow's method **اسئلة على**

$$x = \dots\dots\dots$$

Q3- Use the *Bairstow Method* applying first iteration, to find the roots of the polynomial: (6 marks)

$f(x) = x^3 - 2x^2 + x - 2$. Use $(x^2 - 0.5x + 1)$, as a starting factor.

Given the results of the first synthetic division are:

$$b_3 = 1, b_2 = -1.5, b_1 = -0.75, b_0 = -0.875$$

6.1 ✓
estimate the root

0

Q1

Employ Bairstow's method to determine the first corrected values to original guesses of r and s for the polynomial

$$f_4(x) = 9x^4 + x^3 + 8x - 6$$

Using $x^2 - x + 2$ as starting factor.

اسئلة على اساسيات وعمليات الMatrix

13.5 Q1-Read the questions carefully and choose one correct answer for each question:

(15 marks)

1. If $D = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 7 & 2 \end{bmatrix}$, the value of element d_{22} in the matrix $[D (D^T)]$ is:

a) 11

b) 29

☒ c) 25

d) 53

... within the interval $x_0 = 1$ and $x_1 = 2$, what is the next

4. Matrix which does not have an inverse is classified as:

- a) singular matrix b) non-singular matrix c) linear matrix d) unidentified matrix

6. If $\begin{vmatrix} a & -b & s \\ k & t & r \\ m & g & u \end{vmatrix} = 5$, then the value of $\begin{vmatrix} 3a & -b & s \\ 3k & t & r \\ 3m & g & u \end{vmatrix}$ is:

a) 5

b) 15

c) 30

d) 135

If $|A| = 2$, then the value of $|A^T B|$ is:

7. If A and B are 5×5 matrices with $|A| = 7$ and $|B| = 2$, then the value of $|A+B|$ is

a) 3.5 b) 14 c) 7 d) 0.285

Using the Fixed-Point iteration method by assuming

e.g. Find the determinate of
the given matrix

1) $A = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix} \rightarrow$

2) $B = \begin{bmatrix} 2 & 4 \\ 6 & 12 \end{bmatrix}$

3) $C = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$